Lecture 2

Wilcoxon Signed Rank Sum Test

Quick recap of lecture 1

1. **Parametric techniques** are based on the idea that your data come from a population that has a **known or hypothesized distribution**.
2. **Non-parametric techniques** are used for data that do not belong to a known or hypothesized distribution, they are distribution free, and instead **use ranks, signs** or frequencies of data.
3. **Data types:**
    1. Qualitative (Categorical): Nominal and Ordinal
    2. Quantitative (Numeric): Interval and Ratio
4. **Ranking of data**: order, rank and deal with ties

Learning outcomes for this lecture

1) Know the conditions (type of data, number of samples, relationship between the samples and objective) under which it is appropriate to conduct the WILCOXON SIGNED RANK SUM test

3) Know the assumptions and objective of the WILCOXON SIGNED RANK SUM test and be able to conduct the test for two matched/paired populations

Wilcoxon SIGNED Rank Sum Test

- **When is it appropriate?** *When you want to compare two matched/paired samples of quantitative (interval or ratio-scaled) data with respect to their central locations (typically their medians) i.e. determine whether they come from the same population.*

- Remember: paired t-test we looked at the mean of differences between the samples

- For the Wilcoxon signed rank sum test (the equivalent non-parametric), we look at the **median** (central tendency) of differences

- H_0 : The location of paired differences = 0 / The median difference is equal to zero = 0

- In other words, the samples come from the same population, there is no difference between the samples.

Wilcoxon SIGNED Rank Sum Test

- Alternative hypothesis:

- **Either** H_1 : The median/location of paired difference is ≠ 0 (two-sided test)
- **OR** H_1 : The median/location of paired difference > 0 (one-sided test)
- **OR** H_1 : The median/location of paired difference < 0 (one-sided test)
- Note that you are required to state these hypotheses using information from the question context i.e. do NOT simply state these theoretical hypotheses in a test or exam!

Wilcoxon SIGNED Rank Sum Test

- Data and Assumptions:

  - Two paired samples.
  - Data is quantitative but not normal.
  - Under H0, the distribution of the population of differences within pairs is symmetric around the median.
  - The n paired differences are independent.

Wilcoxon SIGNED Rank Sum Test

- Calculating the test statistic:

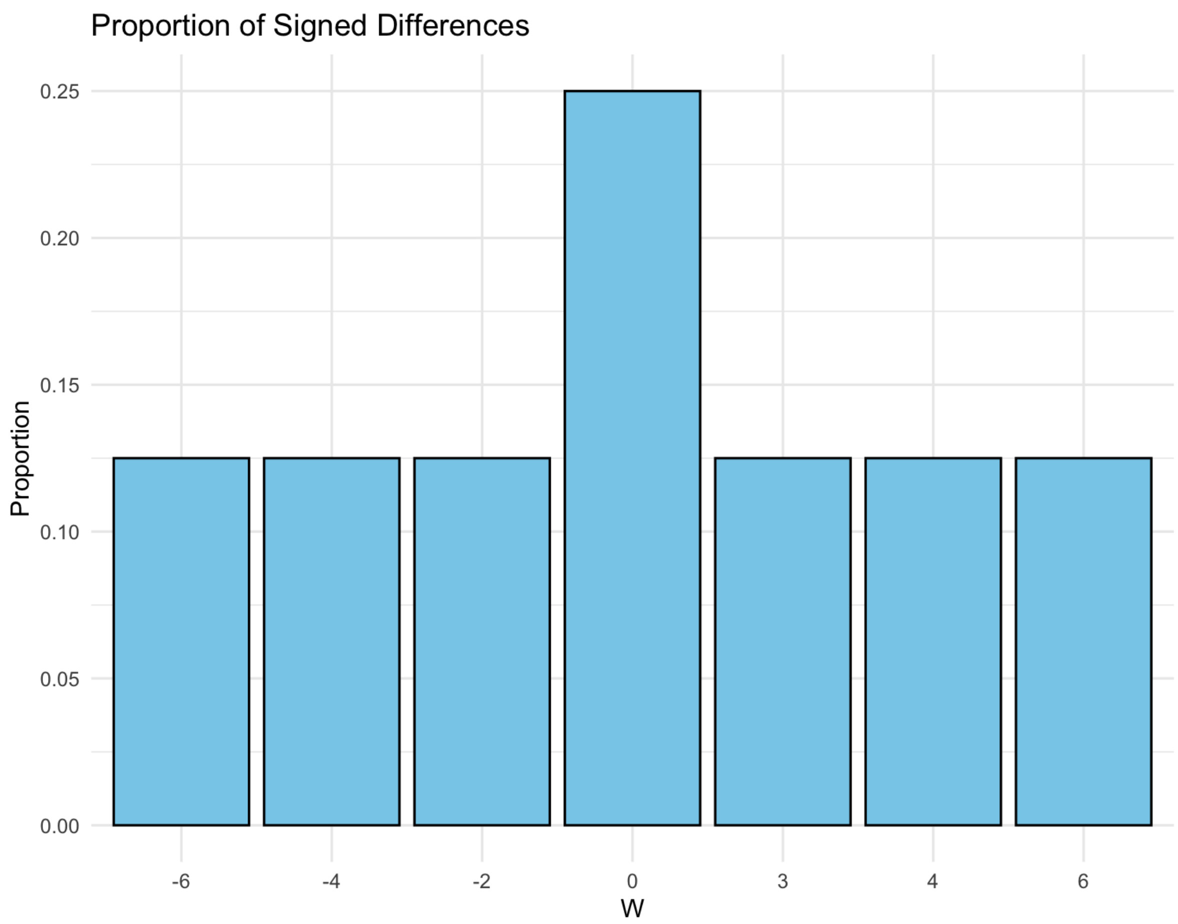
1) Calculate the difference for each pair
2) Eliminate all differences that are equal to 0
3) n = number of non-zero paired differences
4) Record the sign of the paired differences
5) Rank the absolute value of the paired differences
6) W = the sum of the signed ranks of the paired differences

- Note: it is useful to construct a Table to complete steps 1 - 5

Sampling distribution of W

- For small n, it is possible to to deduce the properties of the sampling distribution of W through simple enumeration of all possibilities

Ranks			
1	2	3	W
+	+	+	+6
_	+	+	+4
+	-	+	+3
+	+	_	0
_	_	+	0
_	+	_	-2
+	_	_	-4
_	_	_	-6



Sampling distribution of W

- So for small n, we can get exact critical values and so there is a specific table of critical values to conduct hypothesis test
- BUT it turns out, that for n > 10, the sampling distribution of W starts to become normal!
- $z = \frac{W - \mu_T}{\sigma_T}$ where $\mu_T = 0$ and $\sigma_T = \sqrt{\frac{n(n+1)(2n+1)}{6}}$

- Critical Region:
- Reject H_0 if $|z| \geq z_{\alpha/2}$ (two-sided test) OR
- if $z > z_{\alpha}$ (one-sided > test) OR
- if $z < -z_{\alpha}$ (one-sided < test)
- where z_{α} is the critical value **OR** if the calculated p-value ≤ α
- **(critical value approach)** **(p-value approach)**

Example 1

- Does a “flexi-time” work-schedule help reduce the travel time of workers to work?

- A random sample of 32 workers was selected, and workers recorded their travel time in minutes before and after the program was implemented.

- (Test at the 5% significance level, using the modified (p-value) approach)

Wilcoxon Signed Rank Sum Test --- Example 1

Worker	8:00-Arr	Flextime
1	34	31
2	35	31
3	43	44
4	46	44
5	16	15
6	26	28
7	68	63
8	38	39
9	61	63
10	52	54
11	68	65
12	13	12
13	69	71
14	18	13
15	53	55
16	18	19

Worker	8:00-Arr	Flextime
17	41	38
18	25	23
19	17	14
20	26	21
21	44	40
22	30	33
23	19	18
24	48	51
25	29	33
26	24	21
27	51	50
28	40	38
29	26	22
30	20	19
31	19	21
32	42	38

Wilcoxon Signed Rank Sum Test --- Example 1

- We want to compare two matched/paired samples of quantitative data. The samples do not adhere to a normal distribution, and we are interested in whether there is a difference between them, so we use **the Wilcoxon Signed Rank Sum test.**

- H_0: There is no difference in travel time to work between the normal work-hour and flexi-time work programs. The median difference is zero.

- H_1: Workers take longer to travel to work in the normal work-hour program
- i.e. workers take less time to travel to work in the flexi-time program (hence flexitime does reduce workers' travel time). The median difference is greater than zero.

Example 1

Worker	8:00-Arr	Flextime	Diff	|Diff|
1	34	31	3	3
2	35	31		
3	43	44		
4	46	44		
5	16	15		
6	26	28		
7	68	63		
8	38	39		
9	61	63		
10	52	54		
11	68	65		
12	13	12		
13	69	71		
14	18	13		
15	53	55		
16	18	19		

Worker	8:00-Arr	Flextime	Diff	|Diff|
17	41	38		
18	25	23		
19	17	14		
20	26	21		
21	44	40		
22	30	33		
23	19	18		
24	48	51		
25	29	33		
26	24	21		
27	51	50		
28	40	38		
29	26	22		
30	20	19		
31	19	21		
32	42	38	4	4

Example 1

Worker	8:00-Arr	Flextime	Diff	|Diff|
1	34	31	3	3
2	35	31	4	4
3	43	44	-1	1
4	46	44	2	2
5	16	15	1	1
6	26	28	-2	2
7	68	63	5	5
8	38	39	-1	1
9	61	63	-2	2
10	52	54	-2	2
11	68	65	3	3
12	13	12	1	1
13	69	71	-2	2
14	18	13	5	5
15	53	55	-2	2
16	18	19	-1	1

Worker	8:00-Arr	Flextime	Diff	|Diff|
17	41	38	3	3
18	25	23	2	2
19	17	14	3	3
20	26	21	5	5
21	44	40	4	4
22	30	33	-3	3
23	19	18	1	1
24	48	51	-3	3
25	29	33	-4	4
26	24	21	3	3
27	51	50	1	1
28	40	38	2	2
29	26	22	4	4
30	20	19	1	1
31	19	21	-2	2
32	42	38	4	4

Example 1

Worker	Diff	|Diff|	Rank
1	3	3	
2	4	4	
3	-1	1	
4	2	2	
5	1	1	
6	-2	2	
7	5	5	
8	-1	1	
9	-2	2	
10	-2	2	
11	3	3	
12	1	1	
13	-2	2	
14	5	5	
15	-2	2	
16	-1	1	

Worker	Diff	|Diff|	Rank
17	3	3	
18	2	2	
19	3	3	
20	5	5	
21	4	4	
22	-3	3	
23	1	1	
24	-3	3	
25	-4	4	
26	3	3	
27	1	1	
28	2	2	
29	4	4	
30	1	1	
31	-2	2	
32	4	4	

Example 1

Worker	Diff	|Diff|	Rank
1	3	3	
2	4	4	
3	-1	1	
4	2	2	
5	1	1	
6	-2	2	
7	5	5	
8	-1	1	
9	-2	2	
10	-2	2	
11	3	3	
12	1	1	
13	-2	2	
14	5	5	
15	-2	2	
16	-1	1	

Worker	Diff	|Diff|	Rank
17	3	3	
18	2	2	
19	3	3	
20	5	5	
21	4	4	
22	-3	3	
23	1	1	
24	-3	3	
25	-4	4	
26	3	3	
27	1	1	
28	2	2	
29	4	4	
30	1	1	
31	-2	2	
32	4	4	

8-way tie assign the average of ranks 1 to 8 to each value:

Average =

$$\frac{1+2+3+4+5+6+7+8}{8} = 4.5$$

Example 1

Worker	Diff	|Diff|	Rank
1	3	3	21
2	4	4	27
3	-1	1	4.5
4	2	2	13
5	1	1	4.5
6	-2	2	13
7	5	5	31
8	-1	1	4.5
9	-2	2	13
10	-2	2	13
11	3	3	21
12	1	1	4.5
13	-2	2	13
14	5	5	31
15	-2	2	13
16	-1	1	4.5

Worker	Diff	|Diff|	Rank
17	3	3	21
18	2	2	13
19	3	3	21
20	5	5	31
21	4	4	27
22	-3	3	21
23	1	1	4.5
24	-3	3	21
25	-4	4	27
26	3	3	21
27	1	1	4.5
28	2	2	13
29	4	4	27
30	1	1	4.5
31	-2	2	13
32	4	4	27

Example 1

Worker	Diff	|Diff|	Rank	Signed Rank
1	3	3	21	+21
2	4	4	27	+27
3	-1	1	4.5	-4.5
4	2	2	13	+13
5	1	1	4.5	+4.5
6	-2	2	13	-13
7	5	5	31	+31
8	-1	1	4.5	-4.5
9	-2	2	13	-13
10	-2	2	13	-13
11	3	3	21	+21
12	1	1	4.5	+4.5
13	-2	2	13	-13
14	5	5	31	+31
15	-2	2	13	-13
16	-1	1	4.5	-4.5

Worker	Diff	|Diff|	Rank	Signed Rank
17	3	3	21	+21
18	2	2	13	+13
19	3	3	21	+21
20	5	5	31	+31
21	4	4	27	+27
22	-3	3	21	-21
23	1	1	4.5	+4.5
24	-3	3	21	-21
25	-4	4	27	-27
26	3	3	21	+21
27	1	1	4.5	+4.5
28	2	2	13	+13
29	4	4	27	+27
30	1	1	4.5	+4.5
31	-2	2	13	-13
32	4	4	27	+27

Logic behind the test

- Null hypothesis = median difference is zero / no difference between two paired conditions/measurements
- So, under H0, we would expect distribution of differences between paired observations to be symmetric around zero (median of zero).
- i.e. as many positive differences as negative differences
- Example: Comparing weight of people before and after a weight loss drug. If the drug was not effective, we would expect to see roughly equal number of people who lost weight and gained weight.
- i.e. no consistent pattern in differences
- This test ranks the absolute values of these differences regardless of their sign first and then re-applies the sign afterwards.
- If H0 is true, then the distribution of positive and negative ranks should be random across the ranks
- No systematic bias where larger or smaller ranks are consistently positives or negative = sum of signed ranks (W) should be close to zero!

Logic behind the test

- If larger ranks are more positive, implies that first sample tends to be greater than the second sample and that null hypo is not valid.
- Expect to see W that is greater than zero
- Continuing with the weight loss example, after ranking the differences in weight (ignoring the sign), you'd expect the positive and negative differences to be mixed randomly among the ranks if the drug was ineffective.
- If you consistently saw that all large ranks (indicating large differences) were positive, this might suggest a systematic change (like the diet working), which would contradict the null hypothesis.
- Like before, we ask how far away from zero does W have to be (in either direction for two-sided) to reject the null hypothesis?

Test statistic

n = number of non-zero differences = 32 > 10

Test Statistic: W = sum of ranked signed paired differences = 207

$\Longrightarrow z = \frac{W-\mu_W}{\sigma_W}$ where $\mu_W = 0$ and $\sigma_W = \sqrt{\frac{n(n+1)(2n+1)}{6}} = \sqrt{\frac{32(33)(65)}{6}} = 106.958$

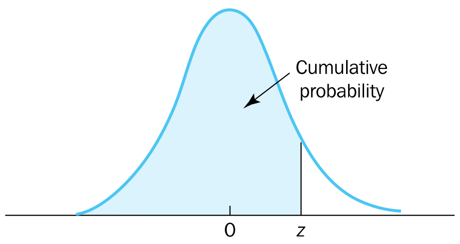
$$\Longrightarrow z = \frac{207 - 0}{106.958} = 1.935$$

Critical Region: Reject H_0 if p-value ≤ 0.05

p-value of test statistic?

Use standard normal distribution table…

TABLE 1 *(Continued)*



Entries in the table give the area under the curve to the left of the z value. For example, for $z = 1.25$, the cumulative probability is .8944.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9913
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9986	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990

Test statistic = 1.935 ≈ 1.94

p-value = 1 – 0.9738 = 0.0262

Conclusion

Conclusion: Since p-value = 0.0262 < 0.05 we Reject H_0 at the 5% significance level and conclude that workers do take longer to travel to work during normal hours than flexitime i.e. there is sufficient evidence that flexi-time does reduce the travel time of workers to work.

Example 2

Context

- Suppose that 16 students are asked two basic probability questions (question A and B) and to give their immediate and intuitive answer without doing any calculations. The instructor (person asking the questions) hypothesizes that if students have developed a good understanding of probability theory, then they will give higher probabilities for question A than B. if they have not listened in class, then their answers will be random and there will be no overall tendency either way.
- H0: The median of the differences between the probabilities given for question A and question B is zero. There is no difference between the probabilities given for question A and B.
- H1: The median of the differences between the probabilities given for question A and question B is greater than zero. The probabilities given for question A are higher than that of question B.

Results

Subj.	X_A	X_B	$X_A—X_B$
1	78	78	0
2	24	24	0
3	64	62	+2
4	45	48	-3
5	64	68	-4
6	52	56	-4
7	30	25	+5
8	50	44	+6
9	64	56	+8
10	50	40	+10
11	78	68	+10
12	22	36	-14
13	84	68	+16
14	40	20	+20
15	90	58	+32
16	72	32	+40
			median difference = +5.5

1	2	3	4	5	6	7
Subject	X_A	X_B	$x_A - x_B$	$\lvert x_A - x_B \rvert$	Abs rank	Signed rank
1	78	78	0	0	---	---
2	24	24	0	0	---	---
3	64	62	+2	2	1	+1
4	45	48	-3	3	2	-2
5	64	68	-4	4	3.5	-3.5
6	52	56	-4	4	3.5	-3.5
7	30	25	+5	5	5	+5
8	50	44	+6	6	6	+6
9	64	56	+8	8	7	+7
10	50	40	+10	10	8.5	+8.5
11	78	68	+10	10	8.5	+8.5
12	22	36	-14	14	10	-10
13	84	68	+16	16	11	+11
14	40	20	+20	20	12	+12
15	90	58	+32	32	13	+13
16	72	32	+40	40	14	+14
						W = 67.0, n = 14

Conclusion

n = number of non-zero differences = 14 > 10

Test Statistic: W = sum of ranked signed paired differences = 67

$\Longrightarrow z = \frac{W-\mu_W}{\sigma_W}$ where $\mu_W = 0$ and $\sigma_W = \sqrt{\frac{n(n+1)(2n+1)}{6}} = \sqrt{\frac{14(15)(29)}{6}} = 31.86$

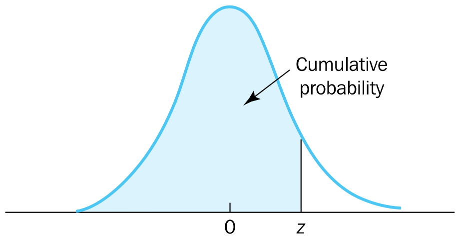
$$\Longrightarrow z = \frac{67-0}{31.86} = 2.10$$

Critical Region: Reject H_0 if p-value ≤ 0.05

p-value of test statistic?

Use standard normal distribution table…

TABLE 1 *(Continued)*



Entries in the table give the area under the curve to the left of the *z* value. For example, for *z* = 1.25, the cumulative probability is .8944.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
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0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
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1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9913
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9986	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990

Test statistic = 2.1

p-value = 1 – 0.9821 = 0.0179

Conclusion

Since p-value = 0.0179 < 0.05 we Reject H_0 at the 5% significance level and conclude that the students : The median of the differences between the probabilities given for question A and question B is greater than zero.

The probabilities given for question A are higher than that of question B / students did give higher probabilities for question A.